

PeerRank: Autonomous LLM Evaluation Through Web-Grounded,
Bias-Controlled Peer Review
Supplementary Material

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A Prompts and constants

This appendix lists the exact prompt templates and constants used for question generation, web-grounded answering, and peer evaluation.

A.1 Constants

Name	Value
Questions per model	NUM_QUESTIONS = 35
Categories	factual knowledge; reasoning / logic; current events; creative / open-ended; practical how-to
Max answer length	MAX_ANSWER_WORDS = 200
Score range	{1,...,10}
Judge reason length	8-20 words
Judge flags	fixed vocabulary array (see Peer Evaluation prompt)

A.2 Prompt: Question Generation

Generate exactly {NUM_QUESTIONS} diverse questions for testing AI capabilities.

Use ONLY these exact category values: {category_list}

Return as JSON object:

```
{"questions": [{"category": "factual knowledge", "question": "Your question"}]}
```

A.3 Prompt: Web-Grounded Answer Generation

Answer this question directly and concisely in {MAX_ANSWER_WORDS} words or less.
Do not start with "Based on..." or similar preambles.

{question}

A.4 Prompt: Peer Evaluation

You are grading responses for a benchmark. Score EACH response independently.

DO NOT try to identify authorship. Ignore writing style and focus on quality.

Scoring rubric (overall 1-10 integer):

- 10: Correct + complete + well-justified; directly answers; no hallucinations.
- 7-9: Mostly correct; minor omissions/imprecision; reasoning mostly sound.
- 4-6: Mixed/partial correctness, unclear reasoning, or misses key constraints.
- 1-3: Mostly incorrect, misleading, evasive, or hallucinated/unsupported.

Priority rules:

- Prioritize correctness/faithfulness over eloquence.
- Penalize confident-sounding unsupported specifics (made-up numbers, names, dates, citations).
- Citations [1][2], source mentions, or "I searched..." phrasing are NEUTRAL - do not reward or penalize.
- If the question is subjective/creative: score instruction-following, coherence, and usefulness; do not mark "incorrect" unless it violates constraints or is nonsensical.

Calibration / score discipline:

- Use the full range when justified; avoid clustering 7-8.
- If there are 3+ responses and quality differs, use at least 3 distinct scores.

```

Question:
{question}

Responses:
{responses}

Output format (STRICT):
- Return ONLY a single JSON object (no markdown, no extra text).
- You MUST include an entry for EVERY label present in Responses, exactly once.
- Each entry MUST contain keys: "score", "reason", "flags" (no other keys).
- "score" MUST be an integer 1--10.
- "reason" MUST be 8--20 words and cite a specific strength or flaw.
- "flags" MUST be an array using only:
  "hallucination", "unsupported_specifics", "evasive", "incorrect",
  "good_uncertainty", "clear_correct"
  Use [] if none apply.

Example:
{"label_example": {"score": 8, "reason": "Correct core claim, minor omission on edge case; clear and grounded.", "flags": ["clear_correct"]}}
```

B Derived metrics example: home advantage

Model	Own Qs	Other Qs	Diff	n_{own}	n_{other}	Cohen's d	Sig
gem-3-pro	8.76	8.34	+0.41	331	3895	+0.23	***
mistral	8.63	8.31	+0.32	378	3920	+0.16	**
deepseek	8.58	8.33	+0.25	350	3875	+0.14	**
gpt-5.2	8.84	8.64	+0.19	371	3854	+0.11	*
kimi-k2.5	8.67	8.48	+0.19	364	3862	+0.10	*
gem-3-flash	7.79	8.03	-0.25	336	3888	-0.14	*
grok-4	7.23	7.70	-0.47	318	3956	-0.24	***

Table A1: Models with statistically significant differences between performance on own-authored vs. other-authored questions (peer scores exclude self-evaluation). Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Average home advantage across models is +0.039 points.

C Category-wise performance

Aggregate rankings can mask specialization by task type. Figure 1 breaks down mean peer scores by question category (shuffle+blind), showing how model performance varies across *Creative*, *Current Events*, *Factual*, *Practical*, and *Reasoning* questions. The category view helps separate broad capability from domain-specific strengths (e.g., open-world retrieval in *Current Events* vs. closed-world performance in *Factual* and *Reasoning*).

The ranking shows a tight top tier (with the top two models nearly tied) and clearer separation toward the lower tier, with non-trivial variance. This suggests the peer evaluation protocol is neither degenerate nor overly compressed. Models from different providers and architectural families interleave, suggesting PEERRANK does not simply reflect shared lineage or training origin. **Provider clustering.** Peer scores nevertheless differ significantly by provider (Kruskal–Wallis $H(8) = 3615.02$, $p < 0.001$; $\eta^2 = 0.071$), indicating a modest provider-level effect.

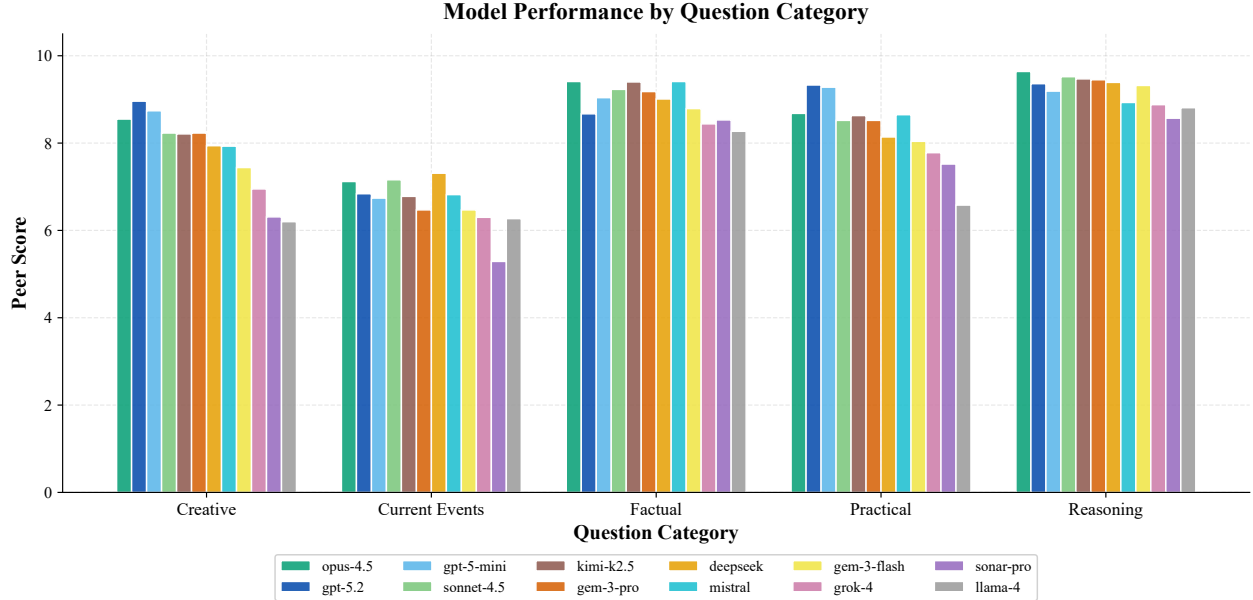


Figure 1: Model performance by question category (shuffle+blind): mean peer score by category.

D Quality-Speed Trade-offs Under Web Grounding

Figure 2 plots each model’s mean peer score against its average wall-clock response time during web-grounded answer generation.

Quality and latency exhibit only a weak correlation. Some models achieve above-average quality with low response times, while others trade substantial latency for modest quality gains. This decoupling suggests that inference speed is not a reliable proxy for answer quality in open-world, web-grounded settings.

Overall, PEERRANK surfaces multi-dimensional performance characteristics—including quality, judgment structure, and efficiency—using a fully endogenous evaluation pipeline without human supervision.

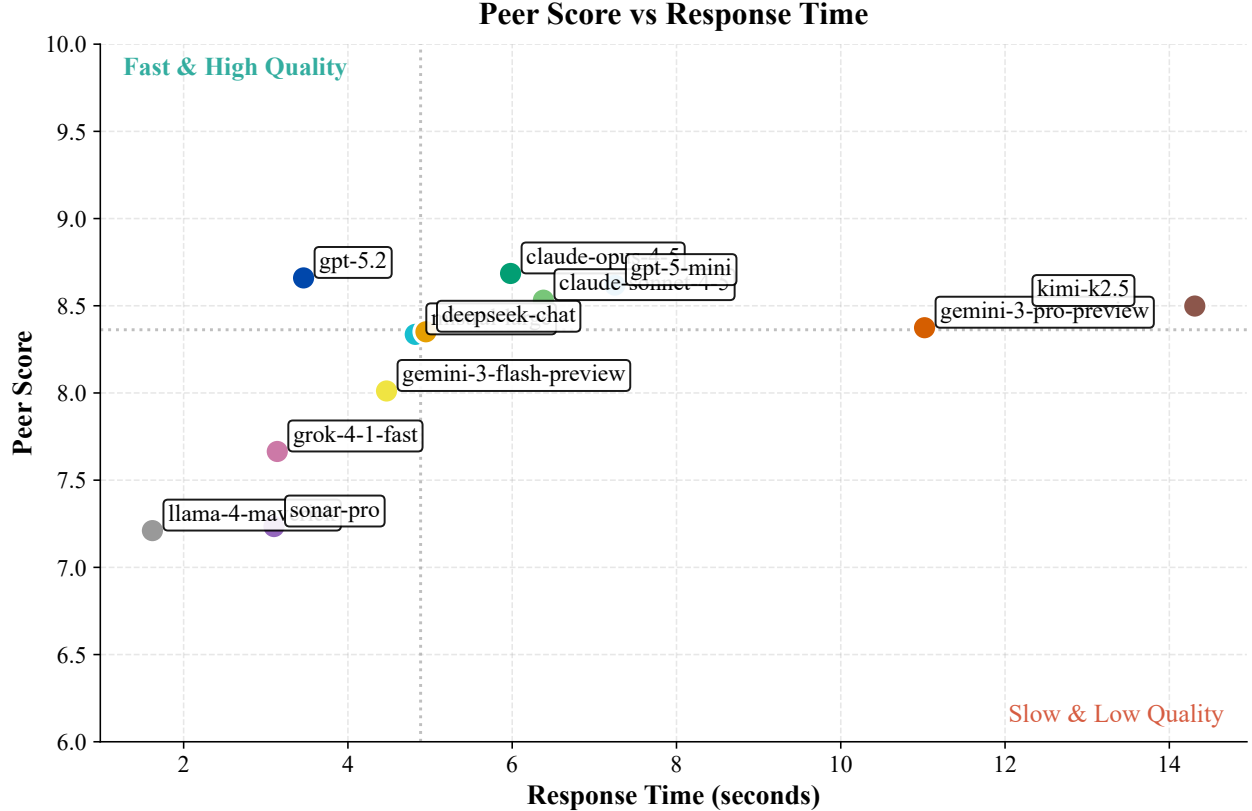


Figure 2: Quality versus speed trade-off under web grounding.

E Judge Generosity and Heterogeneous Evaluation Behavior

We characterize evaluator behavior in two ways: (i) *judge agreement*—how similarly models score the same answers—and (ii) *judge generosity*—the mean score a model assigns to others when acting as an evaluator.

Figure 4 shows that judge generosity is negatively correlated with peer performance across models (Pearson $r = -0.755$, $n = 12$), indicating that higher-ranked models tend to assign lower scores on average. This pattern suggests that evaluator strictness is not merely noise: stronger models may apply tighter standards for correctness, completeness, or rigor when judging peers. At the same time, the correlation is far from perfect, implying that judging style remains partially independent from answer quality—supporting PEERRANK’s design choice to aggregate across many heterogeneous judges rather than relying on a single evaluator model.

Judge generosity varies substantially across models, reflecting differences in calibration and evaluative style. Consistent with the negative generosity–performance correlation (Figure 4, right), higher-performing models tend to judge more harshly on average, though individual deviations remain substantial. This heterogeneity highlights a limitation of single-judge evaluation paradigms and motivates aggregation across multiple, diverse judges as employed in PEERRANK.

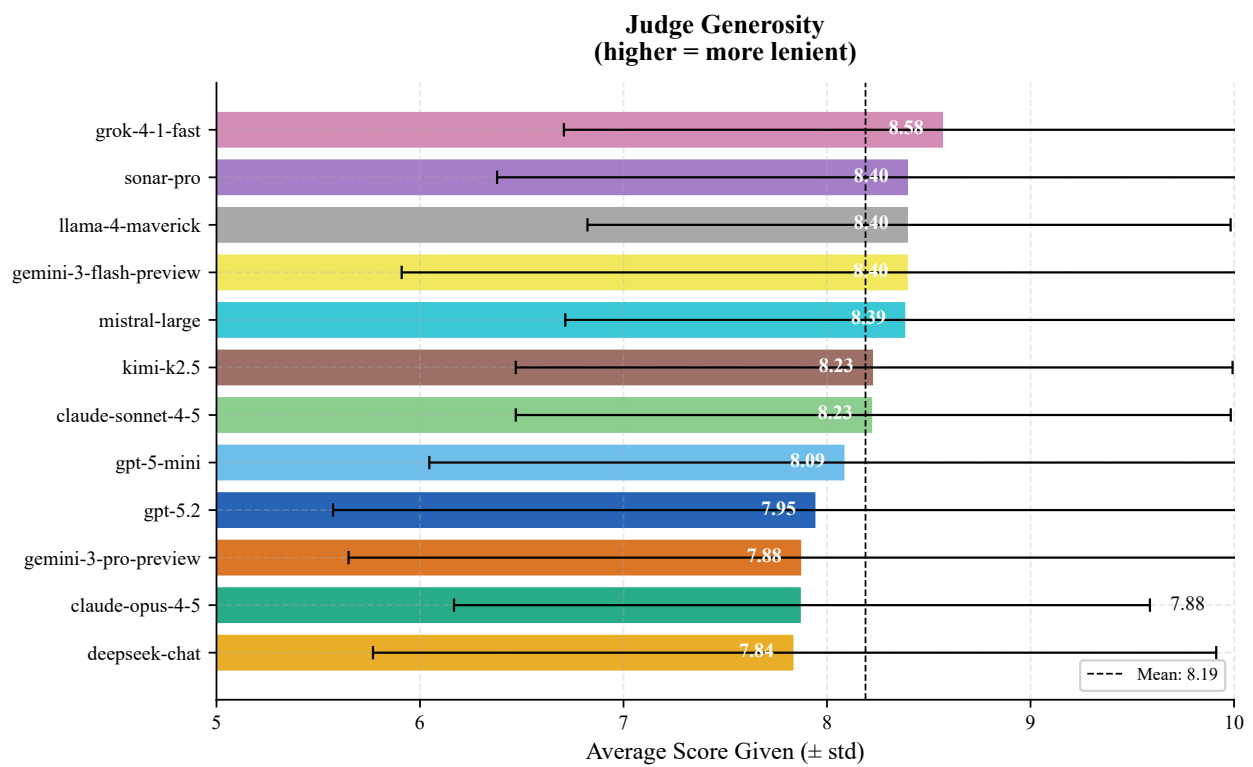


Figure 3: Judge generosity (strictness): mean score each model assigns to peers.

F Task Difficulty and Judge Disagreement

To diagnose sources of model performance limits, we analyze the relationship between question difficulty and judge controversy across categories. We define difficulty as the mean peer score per question across non-self judgments (lower = harder for this cohort), while controversy is the standard deviation of scores (higher = more disagreement). Figure 5 reveals a clear category-based separation:

- **Retrieval drives divergence:** *Current Events* questions (the only category with web grounding enabled) (grey) dominate the "Hard & Controversial" quadrant, suggesting live retrieval is the main differentiator; disagreement likely reflects conflicting retrieval or hallucinations, producing higher score variance.
- **Static reasoning saturation:** *Reasoning/Logic* and *Factual Knowledge* cluster in the "Easy & Consensus" region, suggesting static logic tasks are near-saturated for this cohort and motivating more dynamic, open-world evaluation.

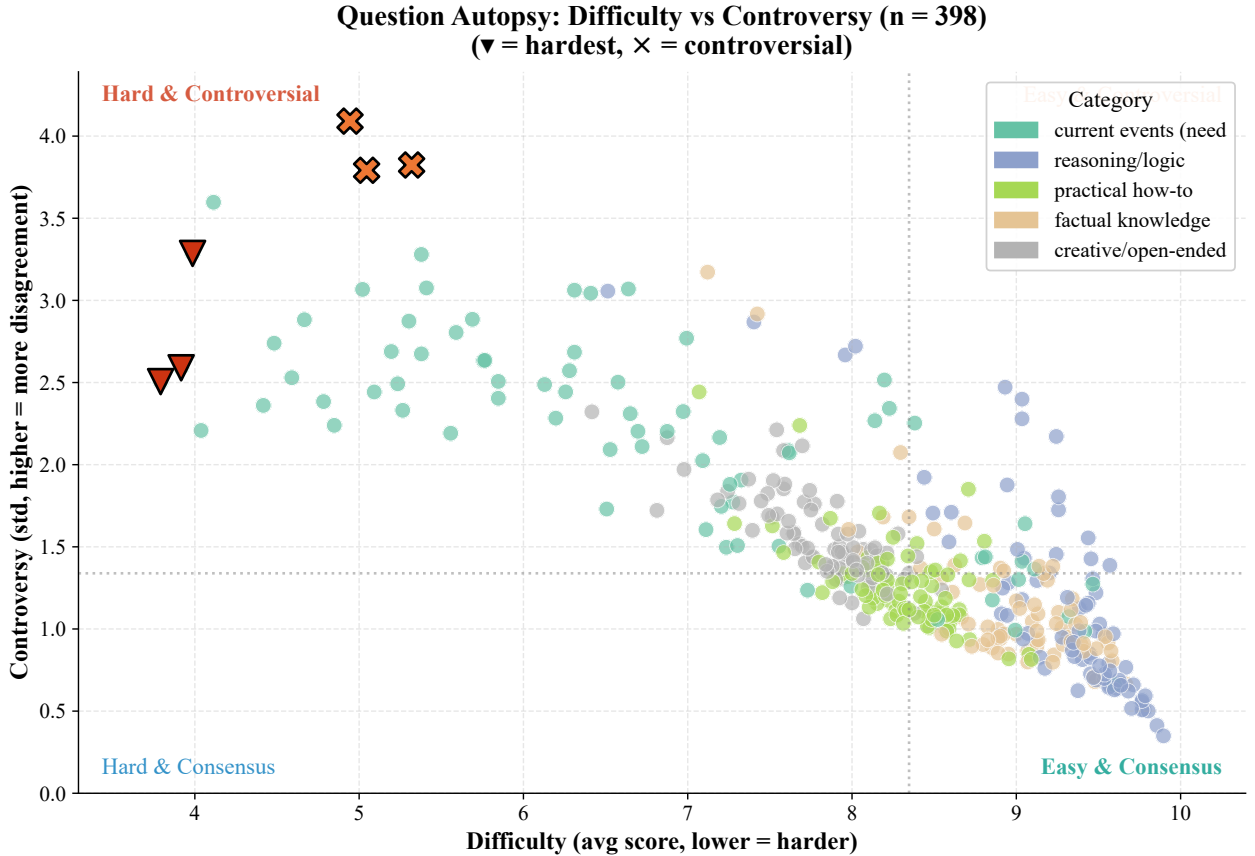


Figure 5: **Question Autopsy: Difficulty vs. Controversy** ($n = 398$); questions with missing evaluations were excluded. **X-axis:** Mean peer score (lower = harder). **Y-axis:** SD of peer scores across judges (higher = more disagreement). *Current Events* concentrate in the hard/controversial quadrant, while *Reasoning* and *Factual Knowledge* cluster in easy/consensus, suggesting saturation on static tasks.

G Reasoning effort as a first-class evaluation dimension

Beyond answer quality and judge effects, we observe substantial *heterogeneity in reasoning allocation* across evaluated models. Using a simple token-efficiency proxy—the character-to-token ratio (chars/token)—we find

that several models likely employ *hidden or budgeted deliberation* during generation. In our cohort, “normal” visible text typically clusters around ~ 2.5 chars/token, whereas ratios below 1.0 strongly suggest that many generated tokens are not surfaced in the final output (e.g., internal chain-of-thought, hidden scratchpads, or provider-side “thinking” budgets).

Table A2 summarizes these modes. Two models (kimi-k2.5, gpt-5-mini) show extremely low ratios on open-ended generation, consistent with heavy internal deliberation, while gemini-3-pro-preview and gemini-3-flash-preview expose an explicit “thinking token” budget. The remaining models exhibit standard visible-text ratios and no evidence of hidden reasoning.

Table A2: Reasoning-mode summary inferred from average GSM8K output statistics (tokens, chars, and chars/token ratio). Ratios < 1.0 indicate substantial non-surfaced generation.

Model	Provider	Avg Tok	Avg Chars	Ratio
kimi-k2.5	Moonshot	1779	253	0.14
gpt-5-mini	OpenAI	1273	209	0.16
gemini-3-pro-preview	Google	1233	444	0.36
gemini-3-flash-preview	Google	896	458	0.51
claude-opus-4-5	Anthropic	165	358	2.17
claude-sonnet-4-5	Anthropic	219	559	2.55
deepseek-chat	DeepSeek	161	433	2.69
gpt-5.2	OpenAI	135	335	2.49
grok-4-1-fast	xAI	120	305	2.55
llama-4-maverick	Together	128	367	2.86
mistral-large	Mistral	149	324	2.18
sonar-pro	Perplexity	156	429	2.74

This heterogeneity plausibly helps explain why “reasoning-heavy” models tend to gain more on structured multi-step tasks than on factual truthfulness. On GSM8K hard questions, the four extended-reasoning models (heavy internal CoT or explicit thinking tokens) achieve a lower mean hard error rate (6%) than standard-mode models (8%), consistent with a deliberation advantage on chained computation. In contrast, on TruthfulQA the advantage is weak: top truthfulness is achieved by standard-mode models (e.g., the Claude family), suggesting that factual truthfulness depends more on *knowledge quality and calibration* (e.g., avoiding common misconceptions) than on deliberation depth per se.

We also observe *task-adaptive reasoning allocation*: for example, gpt-5-mini emits far more tokens on GSM8K than on TruthfulQA ($3.4\times$), indicating that some systems dynamically allocate internal effort based on task structure rather than using a fixed reasoning budget. Taken together, these results motivate treating *reasoning effort* as a first-class evaluation dimension. In PeerRank-style comparisons, differing “deliberation policies” can confound capability measurements unless controlled or explicitly reported. Future work should therefore (i) measure and standardize deliberation budgets across providers where possible, (ii) report reasoning-effort metrics alongside quality metrics, and (iii) test whether rankings are stable under matched-effort constraints.

H Multi-Dimensional Model Characterization

Scalar rankings obscure important trade-offs between model attributes. Figure 6 presents a multi-dimensional comparison across normalized quality, speed, consistency, humility (inverse self bias), and strictness (inverse generosity).

No single model optimizes all dimensions simultaneously. Models with high peer scores may exhibit higher latency or lower humility, while faster models may trade off consistency or strictness. These results caution against over-reliance on single-number leaderboards and motivate richer evaluation artifacts that expose behavioral profiles.

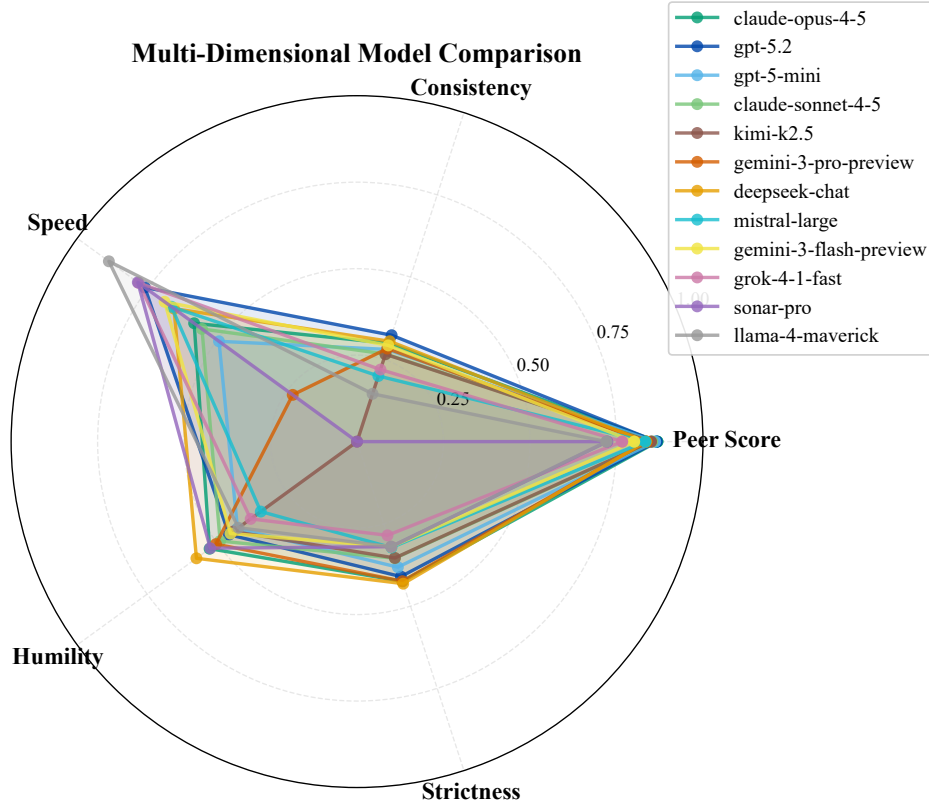


Figure 6: Multi-dimensional comparison of models across normalized evaluation dimensions, including quality, speed, consistency, humility, and strictness. No single model dominates all dimensions.

I Validation on GSM8K

We further test whether peer scores track objective correctness on a structured, exact-match benchmark by running PEERRANK on GSM8K (math reasoning). Each model answers GSM8K questions, we compute accuracy against the gold numeric answers, and rescale accuracy to a 0–10 ground-truth score for comparison with peer scores under shuffle+blind judging.

We evaluated GSM8K on 611 *medium* and *hard* questions. Each model answered all items, we computed exact-match accuracy against gold numeric answers, and rescaled it to a 0–10 ground-truth score ($10 \times$ accuracy) for comparison with shuffle+blind peer scores. Across the 12-model cohort, peer scores strongly correlate with GSM8K ground-truth performance: Pearson $r = 0.873$ ($p = 0.0002$) and Spearman $\rho = 0.763$.

Peer scores track objective math correctness. Peer scores correlate with GSM8K ground-truth performance (See Figure 7), extending external validity beyond factual multiple-choice QA to exact-match math reasoning.

Ceiling-effect caveat (compressed variance). GSM8K accuracy in this cohort is high, compressing the ground-truth range and attenuating linear correlation; nevertheless, rank association remains strong, preserving meaningful ordering.

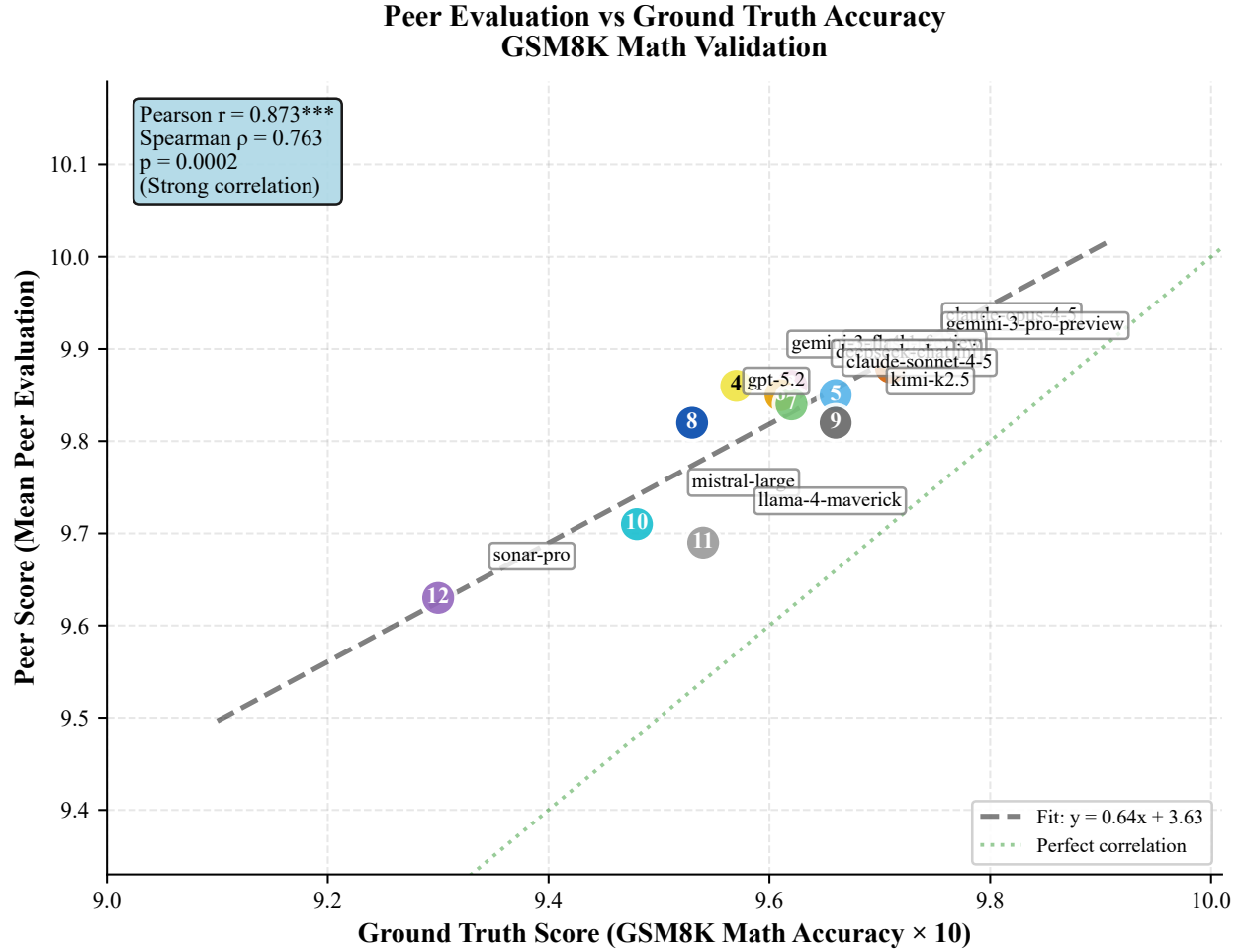


Figure 7: **PeerRank vs. GSM8K ground-truth score.** Peer score (y ; shuffle+blind mean) versus GSM8K ground-truth score (x ; $10 \times$ accuracy).

J Ethics statement

This work evaluates commercial AI systems via public APIs within terms of service. No human subjects were involved. Rankings may influence deployment decisions; readers should consider task-specific requirements beyond aggregate scores.